



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
18-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

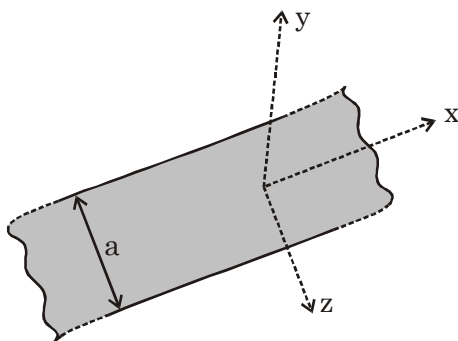
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING**PART-1 : PHYSICS****SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
-
1. Two particles A and B have de-Broglie's wavelengths $30 \text{ \AA}$ and $20 \text{ \AA}$, combined to form a particle C. Momentum is conserved in this process. The possible de-Broglie's wavelength of C is:
- (A) $12 \text{ \AA}$ (B) $20 \text{ \AA}$ (C) $60 \text{ \AA}$ (D) $80 \text{ \AA}$
-

Space for Rough Work

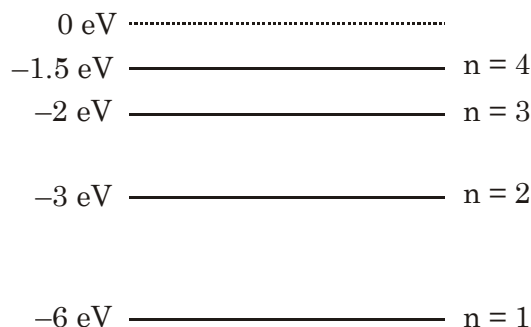
2. A thin metallic plate of uniform mass density is kept along x-axis symmetrically. The length of plate along x-axis is very large as compared to its width 'a' along z-axis. It is given that ' σ ' is mass per unit area. Then choose the correct option(s) :



- (A) The magnitude of gravitational field at $\left(0, \frac{\sqrt{3}a}{2}, 0\right)$ is $\frac{2G\sigma\pi}{3}$.
- (B) The magnitude of gravitational field at $\left(0, \frac{a}{2}, 0\right)$ is $G\sigma\pi$.
- (C) The ratio of magnitude of gravitational field at $\left(0, \frac{a}{2\sqrt{3}}, 0\right)$ and $\left(0, \frac{\sqrt{3}a}{2}, 0\right)$ is 2.
- (D) The ratio of magnitude of gravitational field at $\left(0, \frac{a}{2\sqrt{3}}, 0\right)$ and $\left(0, \frac{a}{2}, 0\right)$ is $\frac{4}{3}$.
3. A long straight staircase consists of steps with horizontal surfaces, each of which has a width $L = 30$ cm and a height $h = 45$ cm. A small ball with a radius of $R = 1$ cm was given a horizontal speed v and angular velocity such that it rolls purely on the top step. The collision with the next step is perfectly inelastic but the ball does not stick to the step. Assume that the ball leaves the step as soon as it reaches the edge of the step. Choose the correct statement(s) :
- (A) The maximum speed v such that it hits the first step below the top is 1 m/s.
- (B) The friction force acting on the ball is always zero.
- (C) If the ball has a speed $v = 30$ cm/s, its speed of rolling on each of the step is 30 cm/s
- (D) If the ball has a speed $v = 60$ cm/s, its horizontal speed after colliding with each step will be progressively lesser.

Space for Rough Work

4. Consider a hypothetical atom, the energy level of atoms are given in the figure. Choose the correct option(s). (Given $hc = 1240 \text{ eV-nm}$)



- (A) An atom in the ground state gets excited by a photon and during de-excitation it emits photons, one of them has wavelength 1240 nm, then the incident photon may have wavelength of 310 nm.
- (B) An atom in the ground state gets excited by a photon and during de-excitation it emits photons, one of them has wavelength 1240 nm, then the incident photon may have wavelength of 1240 nm.
- (C) An atom in the ground state gets excited by a collision with an electron and during de-excitation its emits photons, one of them has wavelength 1240 nm, then incident electron must have an energy of 4 eV.
- (D) An atom in the ground state gets excited by a photon of 8 eV then as a result an ionized atom and an electron of 2eV is obtained.

Space for Rough Work

5. Becquerel performed an experiment in which the particles emitted after radioactive decay were passed through a magnetic field. He measured magnetic rigidity which is a product of magnetic field and the radius of curvature ρ of the particle in the magnetic field. The particles were emitted perpendicular to the magnetic field.

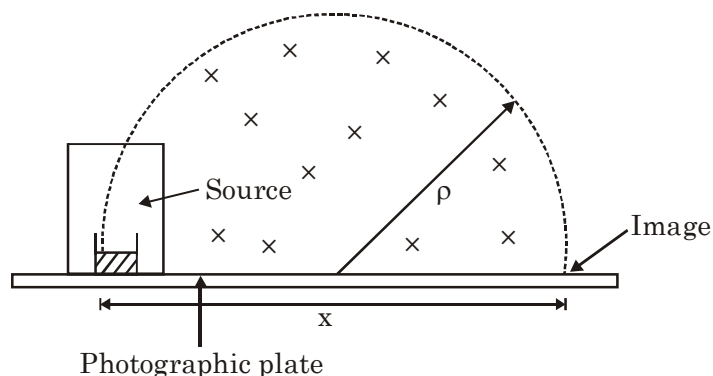
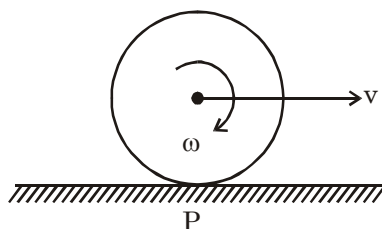


Figure : In Becquerel's arrangement β rays from a radioactive source were deflected in a semicircular arc by a magnetic field into the plane of the paper

Consider the reactions mentioned in options below. In each of the radioactive decay, a statement about the magnetic rigidity is made. Select the option(s) in which the statement correctly describes the outcome of the experiment.

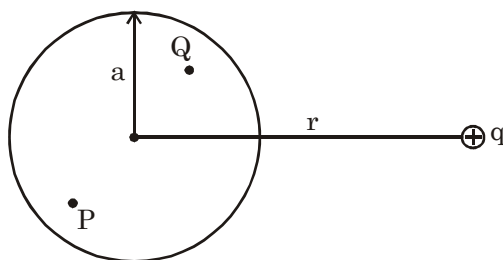
- (A) When $^{37}_{18}\text{Ar}$ decays to $^{37}_{17}\text{Cl}$, the radiation has magnetic rigidity which is found to vary from a certain non-zero minimum value to a certain maximum value
- (B) When $^{238}_{92}\text{U}$ decays to $^{234}_{90}\text{Th}$, the radiation has magnetic rigidity which is the same for all the radiation
- (C) When $^{19}_8\text{O}$ decays to $^{19}_9\text{F}$, the radiation has magnetic rigidity which is found to vary from zero to a certain maximum value
- (D) When $^{25}_{13}\text{Al}$ decays to $^{25}_{12}\text{Mg}$, the radiation has magnetic rigidity which is found to vary from zero to a certain maximum value
6. A body of radius R rolls on a horizontal surface with linear velocity v as shown in figure. Let L_1 and L_2 be the magnitudes of angular momentum of the body about centre of mass and point of contact P . Then find out **INCORRECT** statement(s) :-



- (A) $L_2 = 2L_1$, if radius of gyration $K = R$ (B) $L_2 = 2L_1$, for all cases
- (C) $L_2 > 2L_1$, if radius of gyration $K < R$ (D) $L_2 > 2L_1$, if radius of gyration $K = R$

Space for Rough Work

7. A point charge q is placed outside a conducting spherical shell as shown in figure. There are two points P and Q inside the shell. Select the **CORRECT** option(s) :



- (A) Potential difference between P and Q is zero.
 (B) Electric potential at P and Q due to induced charge on shell is equal.
 (C) Net electric field at P and Q is zero.
 (D) Charge distribution on outer surface of the shell is non-uniform.

8. $^{198}_{79}\text{Au}$ decays to $^{198}_{80}\text{Hg}$ by emitting 2 set of β^- particles and 3 sets of γ rays.

$$m(^{198}\text{Au}) = 197.968233 \text{ amu}$$

$$m(^{198}\text{Hg}) = 197.966760 \text{ amu}$$

$$1 \text{ amu} = 931.5 \text{ MeV}$$

If maximum kinetic energies of β^- particles are 0.284 MeV and 0.960 MeV, what are the possible energies of γ ray photons ?

- (A) 1.088 MeV (B) 0.412 MeV (C) 0.068 MeV (D) 0.676 MeV

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A uniform rod of length L is suspended from its end by a smooth hinge. The rod is free to rotate in

a vertical plane. If the rod is given an angular velocity of $\sqrt{\frac{3g}{L}}$ at the lowest point.

List-I

- (I) Force exerted by the hinge in the horizontal direction
 (II) Force exerted by the hinge in the vertical direction
 (III) Magnitude of Angular velocity of the rod
 (IV) Magnitude of Angular acceleration of the rod

List-II

- (P) increases continuously till the rod stops
 (Q) decreases continuously till the rod stops
 (R) increases and then decreases till the rod stops
 (S) decreases and then increases till the rod stops
 (T) decreases and increases in a periodic manner
 (U) increases and decreases in a periodic manner

- (A) I $\rightarrow$ R,U; II $\rightarrow$ Q,T; III $\rightarrow$ Q,T; IV $\rightarrow$ P,U (B) I $\rightarrow$ P,U; II $\rightarrow$ Q,T; III $\rightarrow$ Q,T; IV $\rightarrow$ P,U
 (C) I $\rightarrow$ R,U; II $\rightarrow$ S,T; III $\rightarrow$ Q,T; IV $\rightarrow$ P,U (D) I $\rightarrow$ P,U; II $\rightarrow$ P,R; III $\rightarrow$ Q,T; IV $\rightarrow$ P,T

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A uniform rod of length L is suspended from its end by a smooth hinge. The rod is free to rotate in a vertical plane. If the rod is given an angular velocity slightly less than $\sqrt{\frac{6g}{L}}$ at the lowest point.

List-I

- (I) Force exerted by the hinge in the horizontal direction
- (II) Force exerted by the hinge in the vertical direction
- (III) Magnitude of Angular velocity of the rod
- (IV) Magnitude of Angular acceleration of the rod

List-II

- (P) increases continuously till the rod stops
- (Q) decreases continuously till the rod stops
- (R) increases and then decreases till the rod stops
- (S) decreases and then increases till the rod stops
- (T) decreases and increases in a periodic manner
- (U) increases and decreases in a periodic manner

- (A) I $\rightarrow$ R,U; II $\rightarrow$ Q,T; III $\rightarrow$ Q,T; IV $\rightarrow$ R,U
- (B) I $\rightarrow$ Q,T; II $\rightarrow$ R,U; III $\rightarrow$ R,U; IV $\rightarrow$ Q,T
- (C) I $\rightarrow$ R,S; II $\rightarrow$ R,S; III $\rightarrow$ P,U; IV $\rightarrow$ Q,T
- (D) I $\rightarrow$ R,U; II $\rightarrow$ S,T; III $\rightarrow$ Q,T; IV $\rightarrow$ R,U

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a hydrogen atom which obeys Bohr's theory and the nucleus is stationary. In deriving the formula for the radius, speed, energy and the wavelength of different lines of spectra, we neglected many phenomena which had a very small contribution. In this question, we are going to consider one of that phenomena. If we also consider the gravitational force of attraction between the nucleus and electron, match the effect that consideration has on the value of the quantity listed in list-I. Assume gravitational as well as electrostatic potential energy to be zero at infinite separation.

List-I

- (I) Radius of orbit
- (II) Speed of electron in its orbit
- (III) Energy of electron in its orbit
- (IV) Wavelength of radiation for excitation to next higher state

List-II

- (P) Same as before
- (Q) Lesser than before
- (R) More than before
- (S) Lesser than before for $n = 1$ and more than before for $n = 2$
- (T) More than before for $n = 1$ and lesser than before for $n = 2$
- (U) Cannot be predicted

(A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ T

(C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ Q

(D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a hydrogen atom which obeys Bohr's theory and the nucleus is stationary. In deriving the formula for the radius, speed, energy and the wavelength of different lines of spectra, we neglected many phenomena which had a very small contribution. In this question, we are going to consider one of that phenomena. The nucleus has magnetic dipole moment which is perpendicular to the electronic orbit. The direction of this magnetic dipole moment is the same as the direction of the angular momentum of electron in the orbit. If we also consider the Magnetic force between the nucleus and electron, match the effect that consideration has on the value of the quantity listed in list-I.

List-I

- (I) Radius of orbit
- (II) Speed of electron in its orbit
- (III) Energy of electron in its orbit
- (IV) Wavelength for excitation to next higher state

List-II

- (P) Same as before
- (Q) Lesser than before
- (R) More than before
- (S) Lesser than before for $n = 1$ and more than before for $n = 2$
- (T) More than before for $n = 1$ and lesser than before for $n = 2$
- (U) Cannot be predicted

(A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

(B) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ S

(C) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ R

(D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ Q

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS. **For Example** : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

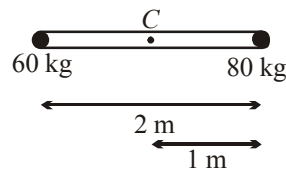
1. A beam of fast (but nonrelativistic) neutrons crosses a beam of slowly moving hydrogen atoms. At what minimum neutron energy (in eV) can hydrogen atoms be ionized? Assume that the mass of neutron is the same as the mass of hydrogen atoms.
2. Consider a β decay reaction

$${}^3_1\text{H} \longrightarrow {}^3_2\text{He} + e^{-1} + \bar{\nu}$$

Atomic mass of ${}^3_1\text{H}$ & ${}^3_2\text{He}$ are 3.016050u & 3.016030u. Find the maximum possible energy of electron (in KeV) :- (Take : 1 amu = 931.5 MeV/c²)
3. Two radioactive samples A_1 and A_2 having half life 3 years and 2 years respectively have been decaying for many years. Today the number of atoms in the sample A_1 is twice the number of atoms in the sample A_2 . Both the samples had same number of atoms X years ago, calculate X.

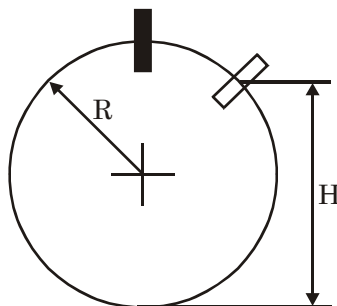
Space for Rough Work

4. A solid sphere has a uniform charge density throughout its volume. The ratio of electric potential at centre to that at surface is given by $\frac{V_{\text{centre}}}{V_{\text{surface}}} = N$. Find N.
5. A uniform rod is mounted so it can rotate about a horizontal axis through the point C as shown in the figure. Two point masses of 60 kg and 80 kg are attached to the ends of the rod as shown. The moment of inertia I of the system (rod and masses) for rotations about C is through an axis perpendicular to the page is 160 kg-m^2 . The rod is initially horizontal, as shown. Note the rod is not massless.



Find the moment of inertia (in kg-m^2) of the system about an axis perpendicular to the rod and passing through its centre of mass.

6. A bead of mass $m = 10 \text{ gm}$ is inserted through a smooth wire ring of radius $R = 6 \text{ cm}$ kept in a vertical plane. The bead is displaced slightly from the equilibrium position, and the bead begins to slide on the ring with an initial velocity equal to zero. What is the force of normal reaction (in N) on the ring at a height of $H = 10 \text{ cm}$?



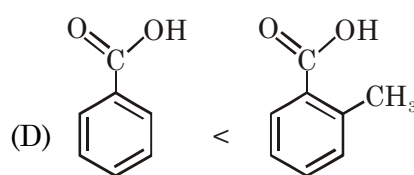
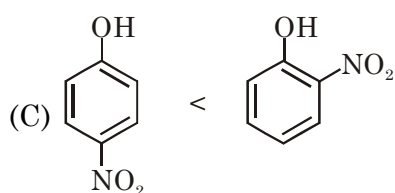
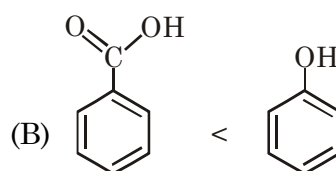
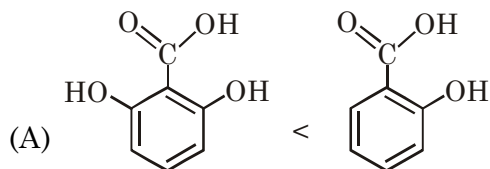
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. In which of the following pair of compounds, order of pK_a is correct :

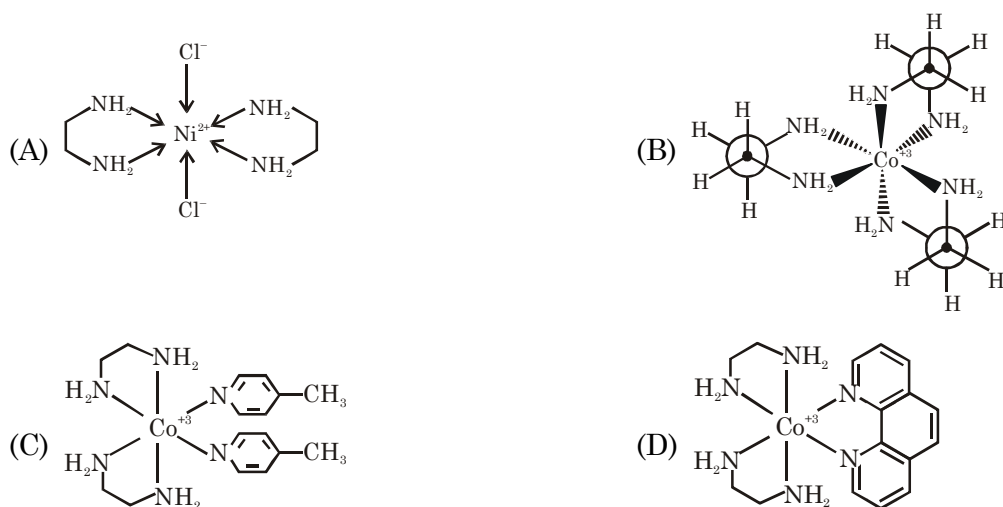


Space for Rough Work

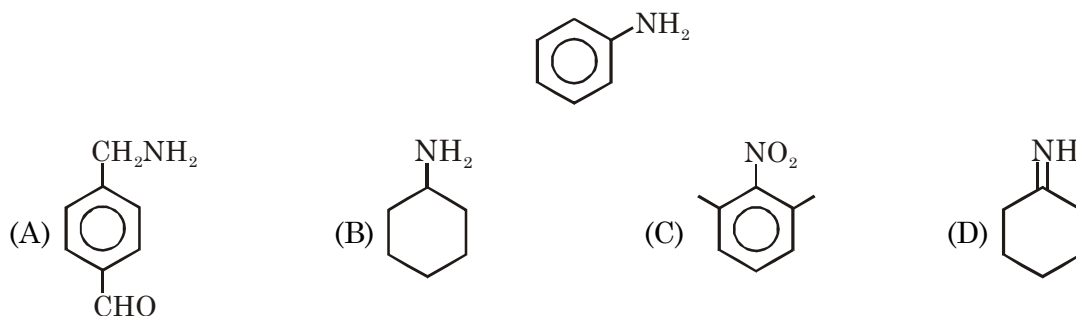
2. 5.6 L of a sample of a mixture of gases containing O_3 , O_2 and He at 1 atm and 273K is reacted with excess KI and liberated I_2 requires 0.5 M, 200 ml standard hypo solution. The same amount of O_2 present in the mixture can be produced by 500 ml $H_2O_2(aq)$ solution on its complete decomposition. 100 ml of this $H_2O_2(aq)$ solution on reaction with excess KI produces I_2 which requires 160 ml of same standard hypo solution. Select the correct options among the following.

- (A) mass % of He in the mixture is $\frac{20}{3}$
 (B) volume strength of H_2O_2 solution is 4.48V corrected at NTP (1 atm, 273 K).
 (C) mole % of O_2 in the mixture is 40
 (D) molarity of H_2O_2 solution is 0.5M.

3. Ethylene diamine (en) is a colorless liquid with ammonia like odour & is a strongly basic amine. Which of the following "en" complexes do not show any geometrical isomerism.

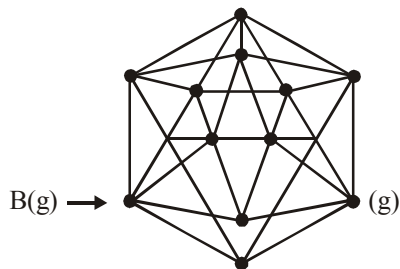


4. Which of the following compound(s) have C–N bond length is more than C–N bond length in Aniline ?



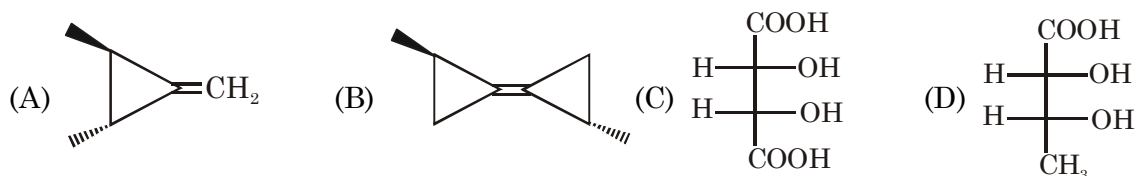
Space for Rough Work

5. Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners and 20 faces) with B atoms at all 12 corners and all bonds are equivalent. Consider the following change:



Given Bond enthalpy of B-B bond = 250 kJmol^{-1} . Select the correct option(s).

- (A) number of B-B bond per atom of B in icosahedral unit is 5
 (B) number of B-B bond per atom of B in icosahedral unit is 2.5
 (C) heat evolved at constant pressure per mole of B atoms undergoing above change is 1250 kJ
 (D) heat evolved at constant pressure per mole of B atoms undergoing above change is 625 kJ
6. Which of the following produces atleast monobasic acid(s) on hydrolysis at room temperature.
 (A) NCl_3 (B) SF_6 (C) ICl (D) XeF_4
7. Which of the following compound is/are chiral :



8. Which of the following statement(s) is/are false?
 (A) All adiabatic processes are isentropic processes.
 (B) If $(dH)_{S,P} < 0$ for a process then the process must be spontaneous.
 (C) $dG = VdP - SdT$ is applicable for closed system considering both PV and non-PV work.
 (D) If a closed system undergoes a reversible process for which $\Delta V = 0$, then the P-V work done on the system in this process must be zero.

Space for Rough Work

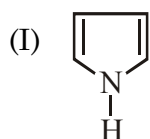
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

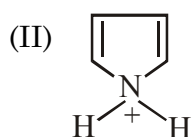
9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List - I

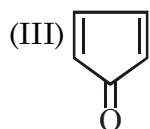
List- II



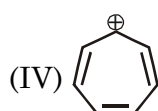
(P) Aromatic



(Q) Non aromatic



(R) Anti-aromatic



(S) Unstable at room temperature because of antiaromatic character

(T) Homocyclic

(U) Hetrocyclic

Which of the following is the only correct combination :

(A) (I), (S)

(B) (III), (P)

(C) (II), (T)

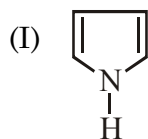
(D) (IV), (P)

Space for Rough Work

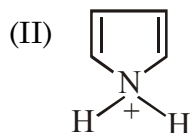
10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List - I

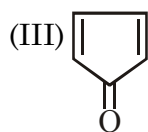
List- II



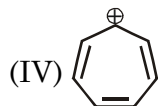
(P) Aromatic



(Q) Non aromatic



(R) Anti-aromatic



(S) Unstable at room temperature because of antiaromatic character

(T) Homocyclic

(U) Hetrocyclic

Which of the following is the only incorrect combination :

(A) (I), (P)

(B) (II), (U)

(C) (III), (S)

(D) (IV), (S)

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List- I (Ligands)		List- II (Characteristics)	
(I)	Thiosulphate ion	(P)	Bidentate ligand
(II)	ortho-phenanthroline	(Q)	Chelating ligand
(III)	oxalate ion	(R)	Contains a chiral centre
(IV)	pn	(S)	Monodentate ligand
		(T)	Ambidentate ligand
		(U)	Show H-bonding in H ₂ O

Which of the following is a correct match.

- (A) I→(P,Q,R) (B) II→(P,Q,T) (C) III→(P,Q,T) (D) IV→(P,Q,U)
12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List- I (Ligands)		List- II (Characteristics)	
(I)	Thiosulphate ion	(P)	Bidentate ligand
(II)	ortho-phenanthroline	(Q)	Chelating ligand
(III)	oxalate ion	(R)	Contains chiral centre(s)
(IV)	pn	(S)	Monodentate ligand
		(T)	Ambidentate ligand
		(U)	Show H-bonding in H ₂ O

Which of the following cases in List-II will match with the ligand "bn".

- (A) P,Q,R,U (B) Q,S,T,U (C) P,R,T,U (D) Q,R,S,T

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

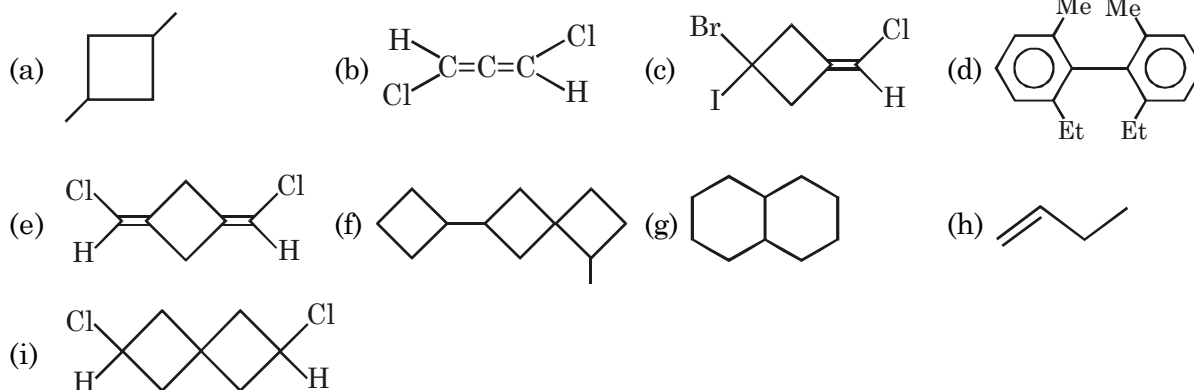
-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- One mole each of benzene ($P_{\text{benzene}}^0 = 90$ torr) toluene ($P_{\text{toluene}}^0 = 45$ torr) and mercury ($P_{\text{mercury}}^0 = 10.25$ torr) are mixed in a well stirred container and allowed to reach equilibrium with their vapours. The vapours are formed in very little amount. Vapour pressure of this mixture in torr is:
- In a 3.27 L closed cylindrical container, the following equilibrium is established at 0°C .
 $2\text{Ag(s)} + 2\text{I}^-(\text{aq}) + 2\text{H}_2\text{O(l)} \rightleftharpoons 2\text{AgI(s)} + \text{H}_2 + 2\text{OH}^-(\text{aq})$.
 The equilibrium constant for the above reaction is 4×10^{-9} . At equilibrium, solution occupies $\frac{1}{3.27}$ th height of container and contains 0.5 M I^- and other species. If gaseous matter weighs 2 gram then calculate $(-\log [\text{OH}^-])$ (Given : $0.083 \times 273 = 22.7$, $R = 0.083$ bar-L/K-mol)

Space for Rough Work

3. How many compounds show geometrical isomerism ?



4. Number of specie(s) that show structural isomerism among the following.

$[\text{Co}(\text{en})_2(\text{NO}_2)\text{Cl}]\text{SCN}$, $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$, $[\text{Pd}(\text{C}_2\text{O}_2\text{S}_2)_2]^{2-}$, Pentaamminenitritocobalt(III)ion, $[\text{Cr}(\text{H}_2\text{O})(\text{NH}_3)_5](\text{NO}_3)_3$, $\text{Ni}(\text{CO})_4$, $\text{K}_2[\text{RuCl}_5\text{N}]$

5. At 300 K, 2463 ml of O_3 (g) dissolves in 360 g of water at 1.0 atm. What mass of ozone in gram can be dissolved in 900 g of water at a pressure of 6 atm and 300K ? ($R = 0.0821 \text{ L-atm / K-mol}$)

6. Number of **CORRECT** order(s) against the mentioned characteristic among the following.

- $\text{O}_2 > \text{O}$ (IE)
- $\text{KCl} < \text{NaCl}$ (Solubility)
- $\text{F} < \text{Cl}$ (E.A.)
- $\text{SOF}_2 < \text{SO}_2\text{F}_2$ ($\text{F} - \hat{\text{S}} - \text{F}$)
- $\text{N}_2^+ > \text{N}_2^-$ (Bond order)
- $\text{FeO} < \text{MnO}_2$ (Acidic strength)
- $\text{AsH}_3 > \text{NH}_3$ (%p character in bond)
- $\text{CH}_3\text{OH} > \text{SiH}_3\text{OH}$ (Acidic strength)

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. If $\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} = \frac{1}{(\theta_1 + \theta_2)}$, then $\frac{d\theta_2}{d\theta_1}$ equals (where x is not dependent on θ_1 or θ_2)
 (A) $\sin^2 x$ (B) $\cos^2 x$ (C) $\tan^2 x$ (D) $\sec^2 x$
2. A right angled triangle has one leg of length 1, another of length 'x' & hypotenuse of length 'y'. The angle opposite to the side x is θ then -
 (A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$ (B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$
 (C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$ (D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist.

Space for Rough Work

3. If $\int \frac{x^6 - x^3 - 2}{x^3 \sqrt{(x^3 + 1)^2 + 3x^4}} dx = f(x) + C$, then -

- (A) $f(1) < f(-1)$ (B) $f(2) > f(3)$ (C) $f(1) > f(-1)$ (D) $f(2) < f(3)$

4. Given $I_n = \int_0^1 (1 - x^2)^n dx$ ($n = 0, 1, 2, \dots$), then which of the following is/are true ?

(A) $\frac{I_{n+1}}{I_n}$ is independent of n .

(B) $\frac{I_{n+1}}{I_n} = \frac{2n+2}{2n+3}$

(C) $I_k = \prod_{r=1}^k \left(\frac{2r}{2r+1} \right), (k \in \mathbb{N})$

(D) there exist some n ($n \in \mathbb{N}$) for which $I_n \geq 1$

5. Which of the following satisfy the equation $\sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1} \tan \sqrt{2-x} - \sin^{-1} \sqrt{\frac{1-x}{2}}$

(A) $\left[2 - \frac{\pi^2}{4}, 1 \right]$

(B) $\left(2 - \frac{\pi^2}{4}, 0 \right)$

(C) $-1, 1$

(D) $\left[1 - \frac{\pi^2}{4}, 1 \right]$

Space for Rough Work

6. Let $f(x)$ be a continuous function in $x \in [0, 5]$ and twice differentiable function in $x \in (0, 5)$ such that $f(4) = f(5) = 0$ then which of the following is/are correct ?
- (A) $\exists$ some ' α ' $\in (0, 5)$ such that $nf(\alpha) + \alpha f'(\alpha) = 0$, $n \in \mathbb{N}$
- (B) $\exists$ some ' α ' $\in (0, 5)$ such that $f''(\alpha) = 0$
- (C) $\exists$ distinct $\alpha_1, \alpha_2 \in (0, 5)$ such that $4f'(\alpha_1) + f'(\alpha_2) + f(0) = 0$
- (D) $\exists$ some ' α ' $\in (0, 4)$ such that $nf(\alpha) + \alpha f'(\alpha) = 0$; $n \in \mathbb{N}$
7. Let $a \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x^5 - 50x^3 + 135x - a$. The equation $f(x) = 0$ will have :
- (A) one real root for $|a| > 216$
- (B) three real roots for $88 < |a| < 216$
- (C) five real roots for $|a| < 88$
- (D) five real roots for $a \in \mathbb{R}$
8. Given f and g are twice differentiable functions and $f' \neq 0$, $g' \neq 0$. If $f(x)g(x) = 1$, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ is equal to :

(A) $\frac{-2g'(x)}{g(x)}$

(B) $\frac{g'(x)}{g(x)}$

(C) $\frac{2f'(x)}{f(x)}$

(D) $\frac{-f'(x)}{f(x)}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I

List-II

(I) $f(x)$ is increasing function in

(P) $\left(0, \frac{3}{2}\right)$

(II) The tangent to $y = f(x)$ at $x = 0$,

meet the curve again at point P,

(Q) $\left(\frac{3}{2}, 4\right)$

whose abscissae lies in

(III) Let N denotes number of real root(s) of equation $f(x) = 0$, then N lies in

(R) (5,7)

(IV) Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is sum of all possible values of a, then k lies in

(S) (8,10)

(T) (9,11)

(U) (11,12)

Which of the following is the only CORRECT combination ?

(A) (I), (S), (T)

(B) (II), (Q)

(C) (III), (P)

(D) (IV), (P)

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I

List-II

(I) $f(x)$ is increasing function in

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meet the curve again at point P,

(Q) $\left(\frac{3}{2}, 4\right)$

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(III) Let N denotes number of real root(s) of equation $f(x) = 0$, then N lies in

(R) (5,7)

(IV) Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is sum of all possible values of a, then k lies in

(S) (8,10)

(T) (9,11)

(U) (11,12)

Which of the following is the only INCORRECT combination ?

(A) (I), (S)

(B) (I), (T)

(C) (II), (R)

(D) (IV), (Q)

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I

List-II

- | | |
|---|-------|
| (I) Points L and M are on the curve $14x^2 - 7xy + y^2 = 2$, each have x co-ordinate '1', point of intersection of tangent at L and M is (h,k), then k is greater than | (P) 0 |
| (II) If $f(x) = x^{\frac{1}{\ln x}}$ then $f'(x) = 0$ for x equals to | (Q) 1 |
| (III) Cosine of the angle of intersection of curves $f(x) = 2^x \ln x$ and $g(x) = x^{2x} - 1$ is | (R) 2 |
| (IV) Consider curve $y^3 - 3xy + 2 = 0$ then total no of horizontal & vertical tangent is | (S) 3 |
| | (T) e |
| | (U) 6 |

Which of the following is the only CORRECT combination ?

- | | |
|----------------|------------------------------|
| (A) (I), (U) | (B) (II), (R), (S), (T), (U) |
| (C) (III), (P) | (D) (IV), (U) |

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I

List-II

- | | |
|---|-------|
| (I) Points L and M are on the curve $14x^2 - 7xy + y^2 = 2$, each have x co-ordinate '1', point of intersection of tangent at L and M is (h,k), then k is greater than | (P) 0 |
| (II) If $f(x) = x^{\frac{1}{\ln x}}$ then $f'(x) = 0$ for x equals to | (Q) 1 |
| (III) Cosine of the angle of intersection of curves $f(x) = 2^x \ln x$ and $g(x) = x^{2x} - 1$ is | (R) 2 |
| (IV) Consider curve $y^3 - 3xy + 2 = 0$ then total no of horizontal & vertical tangent is | (S) 3 |
| | (T) e |
| | (U) 6 |

Which of the following is the only INCORRECT combination ?

- | | | | |
|--------------|--------------|---------------|---------------|
| (A) (I), (T) | (B) (I), (S) | (C) (II), (R) | (D) (IV), (S) |
|--------------|--------------|---------------|---------------|

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If minimum value of $y = \frac{(\sin^{-1} x)^3}{3} + \frac{\pi}{2} (\cos^{-1} x)^2$ exist at $x = \alpha$, then $\cos^{-1} \left(\frac{\cos^{-1}(-\alpha)}{\pi} \right)$ is equal to-
 (Use $\pi = \frac{22}{7}$)
2. The hands of an accurate clock have lengths 3 and 4. If the distance L between the tips of the hands is increasing most rapidly when $L = L_0$, then $(L_0)^2$ is equal to

Space for Rough Work

3. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$, then $|g''(0)|$ is
4. Let $(-1, -1)$ be the focus of a parabola and $3x - y = 8$ be the tangent drawn to it at point $(7, 13)$ on the parabola. If $ax + by + 19 = 0$ is the equation of directrix of the parabola then the value of $\frac{b}{a}$ is (where $a, b \in \mathbb{N}$)
5. If a sequence is such that $a_1 = 1$ and $a_n = n + 1 - a_{n-1} \quad \forall n \geq 2$ and $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{a_1}\right) \left(1 - \frac{1}{a_2}\right) \dots \left(1 - \frac{1}{a_n}\right)$ is _____.
6. For $f(x) = x^4 + |x|$, Let $I_1 = \int_0^\pi f(\cos x) dx$, $I_2 = \int_0^{\pi/2} f(\sin x) dx$, then the value of $\frac{I_1}{I_2}$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator